Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Nelson Dairy

Physical address of dairy:

19481 Geer AVE	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1948

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X210-X016-XXXX

B. OPERATORS

Nelson, Dennis

Operator name: Nelson, Dennis

Telephone no.: (209) 668-4600 (Landline) (Cellular)

19481 Geer AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Nelson, Dennis

Legal owner name: Nelson, Dennis

Telephone no.: (209) 668-4600 (Landline) (Cellular)

19481 Geer AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	168	153	0	0
Maximum number	0	0	168	153	0	0
Average number	0	0	168	153	0	0
Avg live weight (lbs)	0	0	900	750		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 3,174.75 *tons per reporting period*

Total nitrogen from manure: 30,462.90 *lbs per reporting period*

After ammonia losses (30% loss applied): 21,324.03 *lbs per reporting period*

Total phosphorus from manure: 5,155.26 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 680,760 *gallons*

Total nitrogen generated: 1,095.87 *lbs*

Total phosphorus generated: 346.71 *lbs*

Total potassium generated: 4,578.10 *lbs*

Total salt generated: 17,120.49 *lbs*

	680,760	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	680,760	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - DN #3	Ground water
Domestic Well - ND #1	Ground water
Domestic Well - ND #2	Ground water
Irrigation Water - Canal	Surface water
Irrigation Water - ID #987	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	Corral solids *SMI - Souza Brothers*	1,100.00 *ton*	Dry-weight	11.7	20,100.00	7,900.00	22,100.00		0.00

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	39,046.26	15,346.54	42,931.46	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	39,046.26	15,346.54	42,931.46	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Bakers	19	19	2	manure	X045-X210-X063-XXXX
Canal	17	17	2	manure	X045-X210-X016-XXXX
Ethel	50	50	2	manure	X017-X170-X001-XXXX X017-X170-X003-XXXX X017-X170-X011-XXXX
Glantz	37	37	2	manure	X045-X210-X021-XXXX
Lagoon	7	7	2	both	X045-X210-X016-XXXX
Totals for areas that were used for application	130	130	10		
Totals for areas that were not used for application					
Land application area totals	130	130	10		

B. CROPS AND HARVESTS

Bakers

Field name: Bakers

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	246.50 *ton*	Dry-weight		61.6	14,900.00	3,000.00	18,300.00		17.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.97	148.46	29.89	182.34	1,693.84

Bakers

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	527.55 *ton*	As-is		63.5	4,800.00	1,000.00	4,600.00		2.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.77	266.55	55.53	255.45	445.92

Canal

Field name: Canal

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	231.00 *ton*	Dry-weight		66.1	15,700.00	3,200.00	21,300.00		16.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.59	144.64	29.48	196.23	1,520.12

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	419.18 *ton*	As-is		67.9	4,700.00	1,100.00	6,300.00		2.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.66	231.78	54.25	310.69	316.60

Ethel

Field name: Ethel

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	648.30 *ton*	Dry-weight		74.6	22,000.00	4,000.00	30,800.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.97	144.91	26.35	202.87	685.02

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,232.40 *ton*	As-is		64.8	4,800.00	1,000.00	5,100.00		1.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.65	236.62	49.30	251.41	312.34

Glantz

Field name: Glantz

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	455.10 *ton*	Dry-weight		70.5	19,800.00	3,800.00	27,000.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.30	143.69	27.58	195.94	769.24

Glantz

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,202.30 *ton*	As-is		70.9	3,900.00	700.00	5,200.00		1.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.49	253.46	45.49	337.94	302.59

Lagoon

Field name: Lagoon

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	106.30 *ton*	Dry-weight		69.2	16,500.00	3,600.00	24,600.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.19	154.35	33.68	230.12	963.50

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	230.00 *ton*	As-is		68.0	4,000.00	900.00	4,300.00		68.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.86	262.86	59.14	282.57	14,299.43

NUTRIENT BUDGET

A. LAND APPLICATIONS

Bakers - 11/11/2022: Oats, silage-soft dough

Field name: Bakers

Crop: Oats, silage-soft dough — Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.38	3,469,199.00 *gal*
Application event totals		0.00	0.00	0.00	56.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	61.35	3,775,304.00 *gal*
Application event totals		0.00	0.00	0.00	61.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	200.54	72.42	363.94	0.00	210.00 *ton*
Application event totals		200.54	72.42	363.94	0.00	

Bakers - 06/01/2023: Corn, silage

Field name: Bakers

Crop: Corn, silage — Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.38	3,469,199.00 *gal*
Application event totals		0.00	0.00	0.00	56.38	

Bakers - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Import - Souza Bros	Corral solids	280.24	110.14	308.12	0.00	150.00 *ton*
Application event totals		280.24	110.14	308.12	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.06	3,265,128.00 *gal*
Application event totals		0.00	0.00	0.00	53.06	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.74	3,061,058.00 *gal*
Application event totals		0.00	0.00	0.00	49.74	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.43	2,856,987.00 *gal*
Application event totals		0.00	0.00	0.00	46.43	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.74	3,061,058.00 *gal*
Application event totals		0.00	0.00	0.00	49.74	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.06	3,265,128.00 *gal*
Application event totals		0.00	0.00	0.00	53.06	

Canal - 11/11/2022: Oats, silage-soft dough

Canal - 11/11/2022: Oats, silage-soft dough

Field name: Canal

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	58.13	3,200,472.00 *gal*
Application event totals		0.00	0.00	0.00	58.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	61.76	3,400,502.00 *gal*
Application event totals		0.00	0.00	0.00	61.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	192.11	69.37	348.65	0.00	180.00 *ton*
Application event totals		192.11	69.37	348.65	0.00	

Canal - 06/01/2023: Corn, silage

Field name: Canal

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	59.95	3,300,487.00 *gal*
Application event totals		0.00	0.00	0.00	59.95	

Canal - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Import - Souza Bros	Corral solids	313.21	123.10	344.37	0.00	150.00 *ton*
Application event totals		313.21	123.10	344.37	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.50	3,000,443.00 *gal*
Application event totals		0.00	0.00	0.00	54.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.86	2,800,413.00 *gal*
Application event totals		0.00	0.00	0.00	50.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.18	2,652,917.00 *gal*
Application event totals		0.00	0.00	0.00	48.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.89	2,856,987.00 *gal*
Application event totals		0.00	0.00	0.00	51.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	55.60	3,061,058.00 *gal*
Application event totals		0.00	0.00	0.00	55.60	

Ethel - 11/11/2022: Oats, silage-soft dough

Ethel - 11/11/2022: Oats, silage-soft dough

Field name: Ethel

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.41	8,162,820.00 *gal*
Application event totals		0.00	0.00	0.00	50.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	52.09	8,434,914.00 *gal*
Application event totals		0.00	0.00	0.00	52.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	195.96	70.76	355.62	0.00	540.00 *ton*
Application event totals		195.96	70.76	355.62	0.00	

Ethel - 06/01/2023: Corn, silage

Field name: Ethel

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Import - Souza Bros	Corral solids	319.47	125.56	351.26	0.00	450.00 *ton*
Application event totals		319.47	125.56	351.26	0.00	

Ethel - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	52.09	8,434,914.00 *gal*
Application event totals		0.00	0.00	0.00	52.09	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.25	8,298,867.00 *gal*
Application event totals		0.00	0.00	0.00	51.25	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.77	8,707,008.00 *gal*
Application event totals		0.00	0.00	0.00	53.77	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.23	8,134,533.00 *gal*
Application event totals		0.00	0.00	0.00	50.23	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.23	8,134,533.00 *gal*
Application event totals		0.00	0.00	0.00	50.23	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.06	8,267,886.00 *gal*
Application event totals		0.00	0.00	0.00	51.06	

Glantz - 11/11/2022: Oats, silage-soft dough

Glantz - 11/11/2022: Oats, silage-soft dough

Field name: Glantz

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.74	5,600,826.00 *gal*
Application event totals		0.00	0.00	0.00	46.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.41	5,800,856.00 *gal*
Application event totals		0.00	0.00	0.00	48.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	191.25	69.06	347.08	0.00	390.00 *ton*
Application event totals		191.25	69.06	347.08	0.00	

Glantz - 06/01/2023: Corn, silage

Field name: Glantz

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Import - Souza Bros	Corral solids	287.81	113.12	316.45	0.00	300.00 *ton*
Application event totals		287.81	113.12	316.45	0.00	

Glantz - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.42	6,400,944.00 *gal*
Application event totals		0.00	0.00	0.00	53.42	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.25	6,500,959.00 *gal*
Application event totals		0.00	0.00	0.00	54.25	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.42	6,400,944.00 *gal*
Application event totals		0.00	0.00	0.00	53.42	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	53.49	6,410,200.00 *gal*
Application event totals		0.00	0.00	0.00	53.49	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.20	6,734,327.00 *gal*
Application event totals		0.00	0.00	0.00	56.20	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	57.90	6,938,397.00 *gal*
Application event totals		0.00	0.00	0.00	57.90	

Lagoon - 11/11/2022: Oats, silage-soft dough

Lagoon - 11/11/2022: Oats, silage-soft dough

Field name: Lagoon

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.80	12.78	145.33	598.26	122,400.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.45	1,030,455.00 *gal*
Application event totals		38.80	12.78	145.33	643.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	123.55	75.50	260.83	0.00	65.00 *ton*
Application event totals		123.55	75.50	260.83	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.19	14.89	169.27	696.80	142,560.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.45	1,030,455.00 *gal*
Application event totals		45.19	14.89	169.27	742.26	

Lagoon - 06/01/2023: Corn, silage

Field name: Lagoon

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Import - Souza Bros	Corral solids	253.55	99.65	278.78	0.00	50.00 *ton*
Application event totals		253.55	99.65	278.78	0.00	

Lagoon - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.12	1,000,148.00 *gal*
Application event totals		0.00	0.00	0.00	44.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.29	800,118.00 *gal*
Application event totals		0.00	0.00	0.00	35.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.70	900,133.00 *gal*
Application event totals		0.00	0.00	0.00	39.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.40	5.40	46.60	212.44	118,800.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.12	1,000,148.00 *gal*
Application event totals		13.40	5.40	46.60	256.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.62	7.41	131.77	422.22	133,650.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	61.76	1,400,207.00 *gal*
Application event totals		26.62	7.41	131.77	483.99	

Lagoon - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.54	9.06	161.05	516.05	163,350.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	57.35	1,300,192.00 *gal*
Application event totals		32.54	9.06	161.05	573.40	

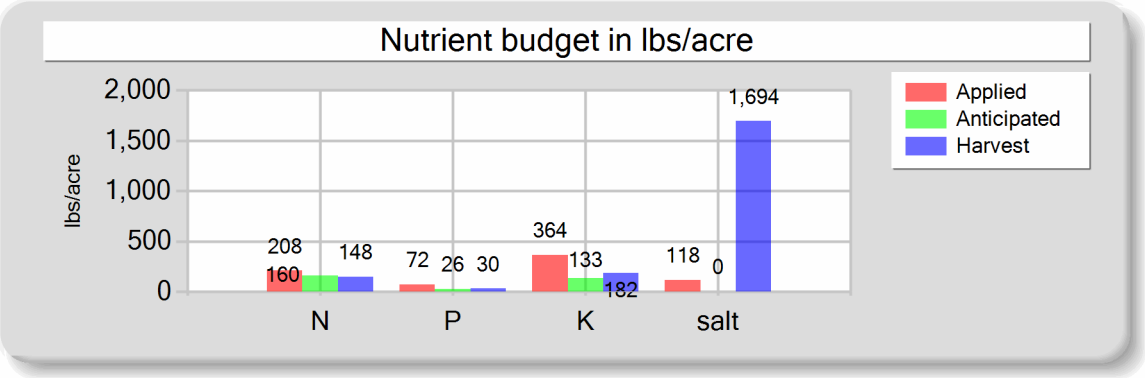
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	52.94	1,200,177.00 *gal*
Application event totals		0.00	0.00	0.00	52.94	

B. NUTRIENT BUDGET

Bakers - 11/11/2022: Oats, silage-soft dough

Field name: Bakers | Crop: Oats, silage-soft dough | Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	200.54	72.42	363.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	117.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.54	72.42	363.94	117.73
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	148.46	29.89	182.34	1,693.84
Nutrient balance	59.08	42.53	181.60	-1,576.12
Applied to removed ratio	1.40	2.42	2.00	0.07

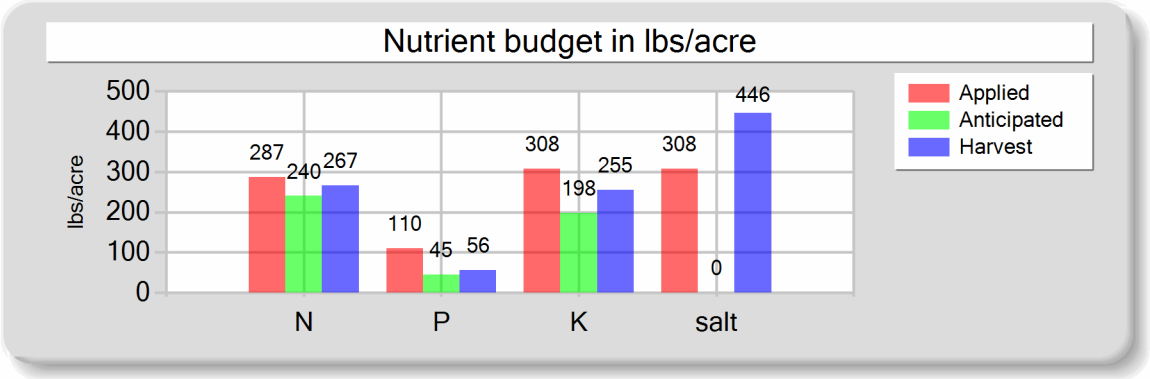
Fresh water applied	
7,244,503.00	*gallons*
266.79	*acre-inches*
14.04	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bakers - 06/01/2023: Corn, silage

Field name: Bakers | Crop: Corn, silage | Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	280.24	110.14	308.12	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	308.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	287.24	110.14	308.12	308.42
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	266.55	55.53	255.45	445.92
Nutrient balance	20.68	54.61	52.68	-137.50
Applied to removed ratio	1.08	1.98	1.21	0.69

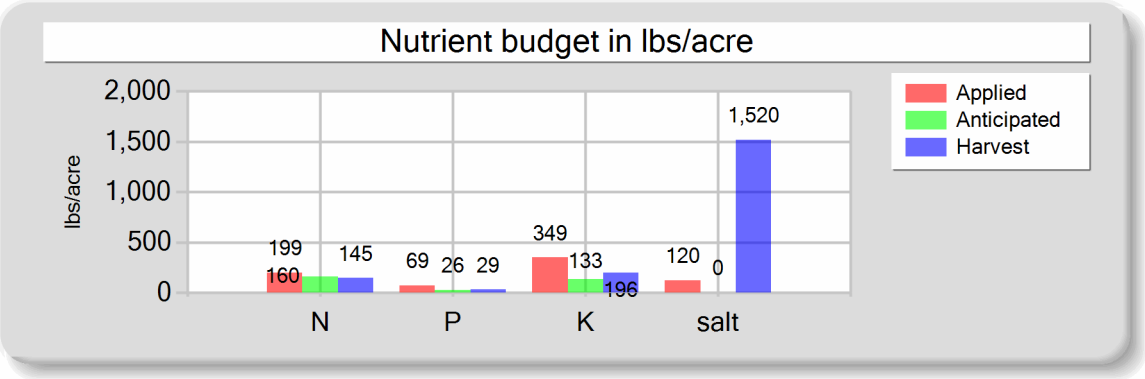
Fresh water applied	
18,978,558.00	*gallons*
698.92	*acre-inches*
36.79	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Canal - 11/11/2022: Oats, silage-soft dough

Field name: Canal | Crop: Oats, silage-soft dough | Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	192.11	69.37	348.65	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	119.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	199.11	69.37	348.65	119.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	144.64	29.48	196.23	1,520.12
Nutrient balance	54.47	39.89	152.42	-1,400.22
Applied to removed ratio	1.38	2.35	1.78	0.08

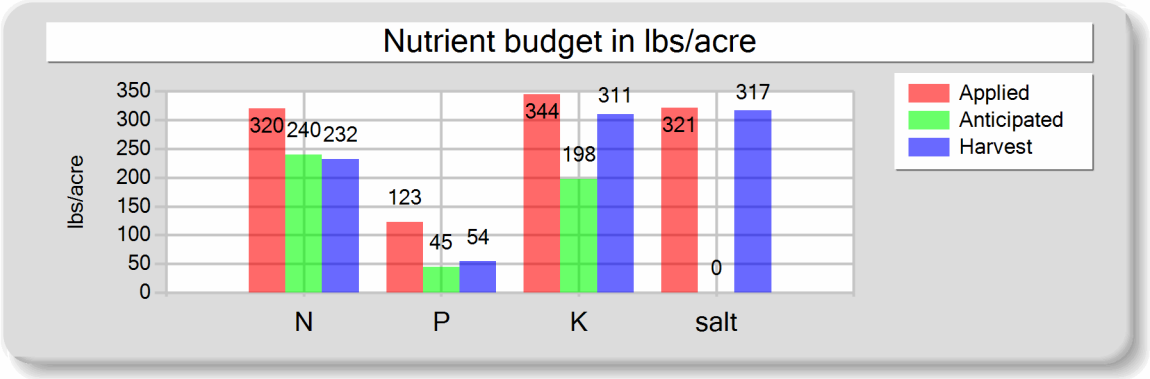
Fresh water applied	
6,600,974.00	*gallons*
243.09	*acre-inches*
14.30	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Canal - 06/01/2023: Corn, silage

Field name: Canal | Crop: Corn, silage | Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	313.21	123.10	344.37	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	320.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	320.21	123.10	344.37	320.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	231.78	54.25	310.69	316.60
Nutrient balance	88.42	68.85	33.68	4.37
Applied to removed ratio	1.38	2.27	1.11	1.01

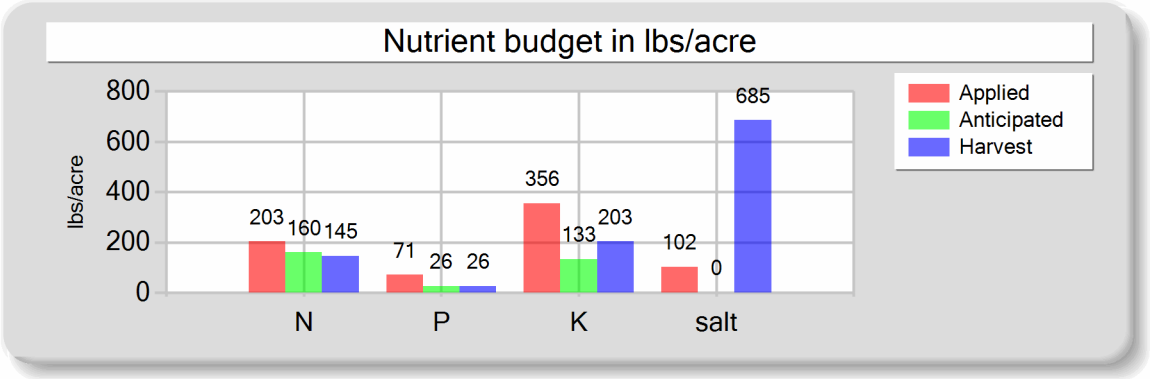
Fresh water applied	
17,672,305.00	*gallons*
650.81	*acre-inches*
38.28	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ethel - 11/11/2022: Oats, silage-soft dough

Field name: Ethel | Crop: Oats, silage-soft dough | Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	195.96	70.76	355.62	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	102.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.96	70.76	355.62	102.50
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	144.91	26.35	202.87	685.02
Nutrient balance	58.05	44.41	152.75	-582.52
Applied to removed ratio	1.40	2.69	1.75	0.15

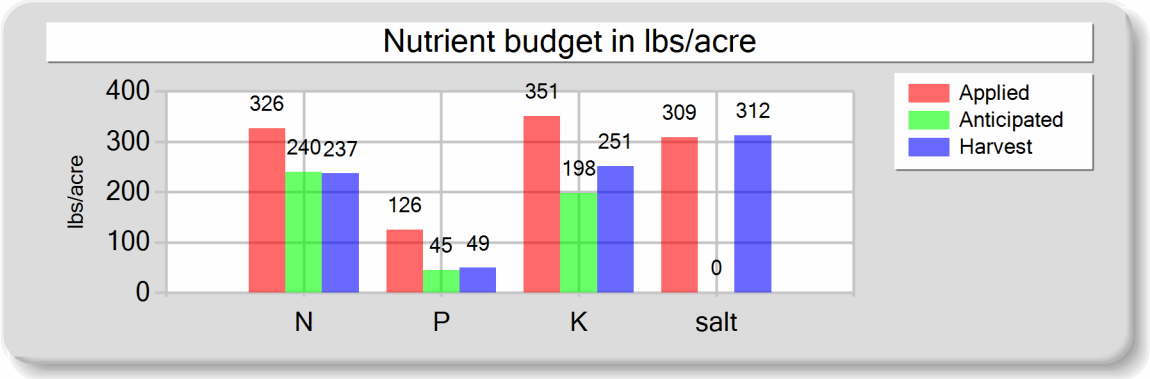
Fresh water applied	
16,597,734.00	*gallons*
611.24	*acre-inches*
12.22	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ethel - 06/01/2023: Corn, silage

Field name: Ethel | Crop: Corn, silage | Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	319.47	125.56	351.26	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	308.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	326.47	125.56	351.26	308.63
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.62	49.30	251.41	312.34
Nutrient balance	89.85	76.27	99.85	-3.71
Applied to removed ratio	1.38	2.55	1.40	0.99

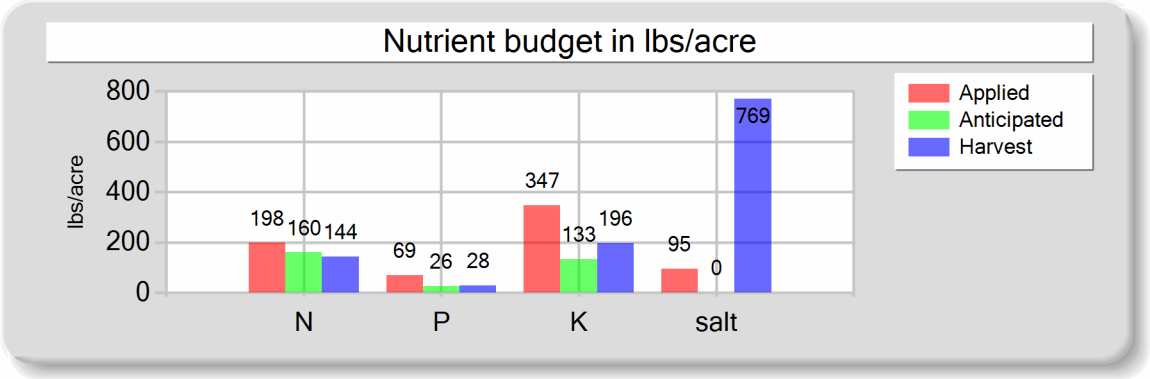
Fresh water applied	
49,977,741.00	*gallons*
1,840.51	*acre-inches*
36.81	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Glantz - 11/11/2022: Oats, silage-soft dough

Field name: Glantz | Crop: Oats, silage-soft dough | Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	191.25	69.06	347.08	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	95.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.25	69.06	347.08	95.15
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	143.69	27.58	195.94	769.24
Nutrient balance	54.56	41.49	151.14	-674.09
Applied to removed ratio	1.38	2.50	1.77	0.12

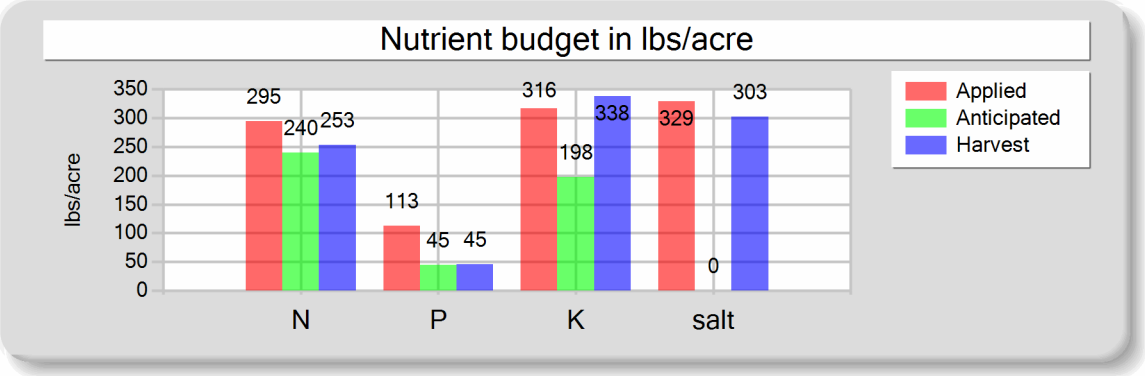
Fresh water applied	
11,401,682.00	*gallons*
419.89	*acre-inches*
11.35	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Glantz - 06/01/2023: Corn, silage

Field name: Glantz | Crop: Corn, silage | Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	287.81	113.12	316.45	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	328.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.81	113.12	316.45	328.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	253.46	45.49	337.94	302.59
Nutrient balance	41.35	67.63	-21.50	26.08
Applied to removed ratio	1.16	2.49	0.94	1.09

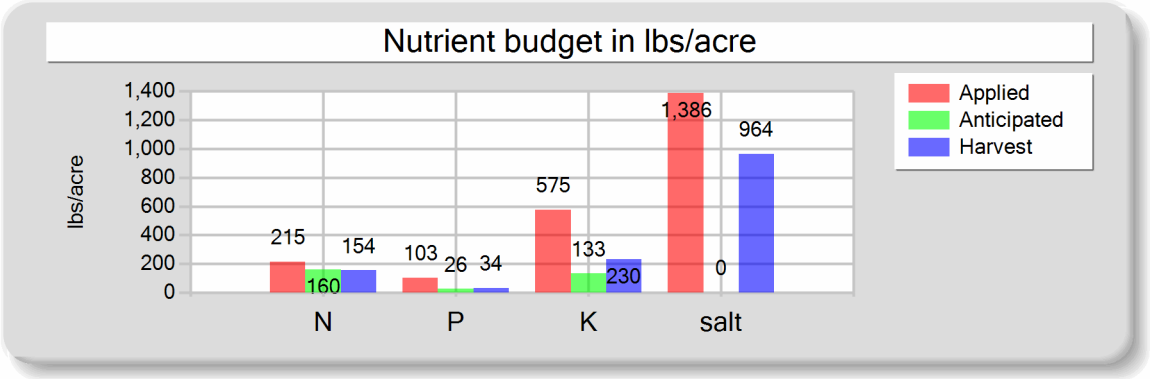
Fresh water applied	
39,385,771.00	*gallons*
1,450.44	*acre-inches*
39.20	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lagoon - 11/11/2022: Oats, silage-soft dough

Field name: Lagoon | Crop: Oats, silage-soft dough | Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	123.55	75.50	260.83	0.00
Process wastewater	83.99	27.67	314.61	1,295.07
Fresh water	0.00	0.00	0.00	90.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.54	103.17	575.44	1,385.97
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.35	33.68	230.12	963.50
Nutrient balance	60.19	69.50	345.32	422.47
Applied to removed ratio	1.39	3.06	2.50	1.44

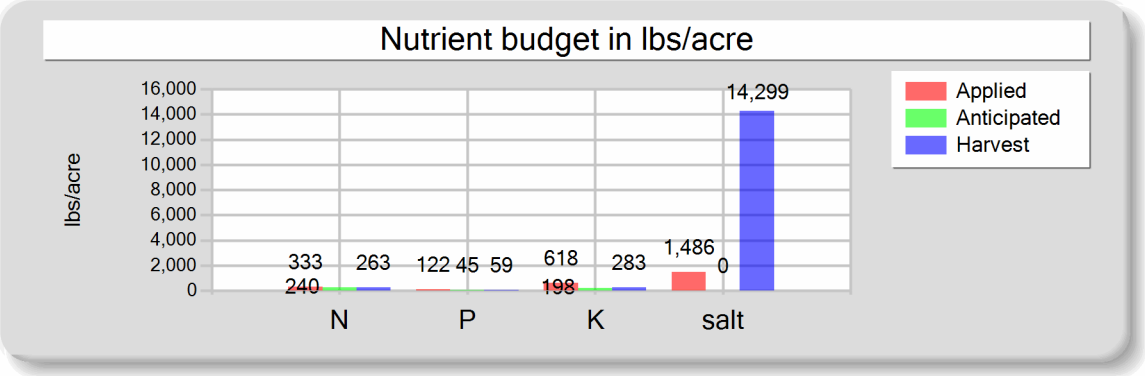
Fresh water applied	
2,060,910.00	*gallons*
75.90	*acre-inches*
10.84	*inches/acre*

Process wastewater applied	
264,960.00	*gallons*
9.76	*acre-inches*
1.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lagoon - 06/01/2023: Corn, silage

Field name: Lagoon Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	253.55	99.65	278.78	0.00
Process wastewater	72.56	21.86	339.41	1,150.72
Fresh water	0.00	0.00	0.00	335.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	333.11	121.51	618.18	1,486.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	262.86	59.14	282.57	14,299.43
Nutrient balance	70.25	62.37	335.61	-12,813.43
Applied to removed ratio	1.27	2.05	2.19	0.10

Fresh water applied	
7,601,123.00	*gallons*
279.92	*acre-inches*
39.99	*inches/acre*

Process wastewater applied	
415,800.00	*gallons*
15.31	*acre-inches*
2.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

SMI - Souza Brothers

Sample and source description: SMI - Souza Brothers
Sample date: 05/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 11.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	7,900.00	22,100.00							0.00
DL	100.00	100.00	100.00							1.00

Solid Manure

Sample and source description: Solid Manure
Sample date: 05/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 7.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,200.00	4,400.00	15,200.00							0.00
DL	100.00	100.00	100.00							1.00

Solid Manure

Sample and source description: Solid Manure
Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 16.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	3,900.00	19,600.00							0.00
DL	100.00	100.00	100.00							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	265.00	106.00	0.00	0.90	87.60	996.00								5,490.00	4,100
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon water

Sample and source description: Lagoon water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	166.00	85.80	0.00	1.10	46.50	827.00								4,880.00	2,650
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	94.30	42.90	0.00	0.30	38.10	329.00								1,950.00	1,500
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	95.40	35.40	0.00	0.00	36.00	332.00								1,870.00	1,550
DL	1.00	0.50	1.00	1.00	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - DN #3

DN #3

Sample description: DN #3

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	15.90	0.00	15.90								509.00	0
DL	1.00	1.00	0.10								10.00	1

Domestic Well - ND #1

ND #1

Sample description: ND #1

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.80	0.00	5.80								473.00	0
DL	1.00	1.00	0.10								10.00	1

Domestic Well - ND #2

ND #2

Sample description: ND #2

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.90	0.00	5.90								474.00	0
DL	1.00	1.00	0.10								10.00	1

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								31.10	37
DL	1.00	1.00	1.00								10.00	10

Irrigation Water - ID #987

ID #987

Sample description: ID #987

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	12.10	0.00	12.10								461.00	332
DL	1.00	1.00	0.10								10.00	10

D. SOIL ANALYSES

Bakers

Bakers

Sample and source description: Bakers

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			51.00				
DL			2.00				

Canal

Canal-Triangle

Sample and source description: Canal-Triangle

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			95.00				
DL			2.00				

Ethel

Ethel-Dennis

Sample and source description: Ethel-Dennis

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			113.00				
DL			2.00				

Glantz

Glantz-Darryl

Sample and source description: Glantz-Darryl

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			64.00				
DL			2.00				

Lagoon

Lagoon

Lagoon

Sample and source description: Lagoon

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			36.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Bakers - 11/11/2022: Oats, silage-soft dough

Bakers

Sample and source description: Bakers

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	3,000.00	18,300.00		17.00
DL	100.00	100.00	100.00		0.01

Bakers - 06/01/2023: Corn, silage

Field #20 Baker

Sample and source description: Field #20 Baker

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,000.00	4,600.00		2.20
DL	100.00	100.00	100.00		0.10

Canal - 11/11/2022: Oats, silage-soft dough

Canal - Triangle

Sample and source description: Canal - Triangle

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	3,200.00	21,300.00		16.50
DL	100.00	100.00	100.00		0.01

Canal - 06/01/2023: Corn, silage

Field #18 Triangle

Sample and source description: Field #18 Triangle

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	1,100.00	6,300.00		2.00
DL	100.00	100.00	100.00		0.10

Ethel - 11/11/2022: Oats, silage-soft dough

Ethel - Dennis A

Sample and source description: Ethel - Dennis A

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	4,000.00	30,800.00		10.40
DL	100.00	100.00	100.00		0.01

Ethel - 11/11/2022: Oats, silage-soft dough

Ethel - Dennis B

Sample and source description: Ethel - Dennis B

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	3,900.00	30,600.00		9.90
DL	100.00	100.00	100.00		0.01

Ethel - 06/01/2023: Corn, silage

Field #17 Dennis 1

Sample and source description: Field #17 Dennis 1

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,000.00	5,100.00		1.80
DL	100.00	100.00	100.00		0.10

Field #17 Dennis 2

Sample and source description: Field #17 Dennis 2

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	1,000.00	5,100.00		64.60
DL	100.00	100.00	100.00		0.01

Glantz - 11/11/2022: Oats, silage-soft dough

Glantz - 11/11/2022: Oats, silage-soft dough

Glantz - Darryl

Sample and source description: Glantz - Darryl

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	3,800.00	27,000.00		10.60
DL	100.00	100.00	100.00		0.01

Glantz - 06/01/2023: Corn, silage

Field #16 Darrels

Sample and source description: Field #16 Darrels

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	700.00	5,200.00		1.60
DL	100.00	100.00	100.00		0.10

Lagoon - 11/11/2022: Oats, silage-soft dough

Lagoon

Sample and source description: Lagoon

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	3,600.00	24,600.00		10.30
DL	100.00	100.00	100.00		0.01

Lagoon - 06/01/2023: Corn, silage

Field #19 Dennis Lagoon

Sample and source description: Field #19 Dennis Lagoon

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	900.00	4,300.00		68.00
DL	100.00	100.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

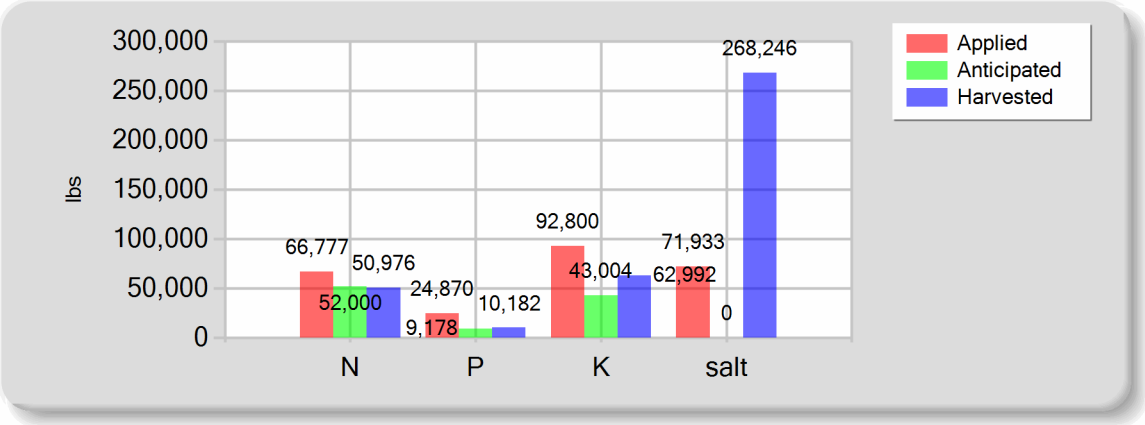
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

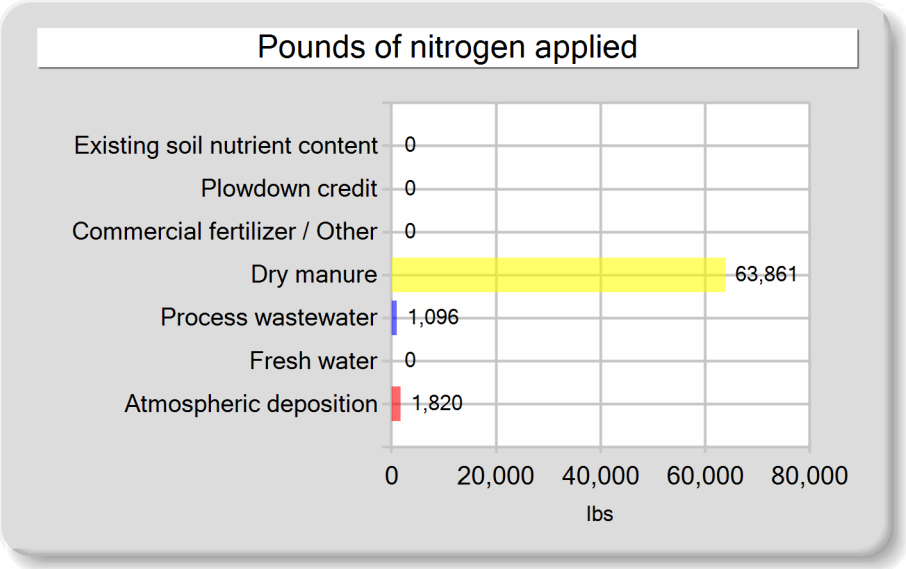
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

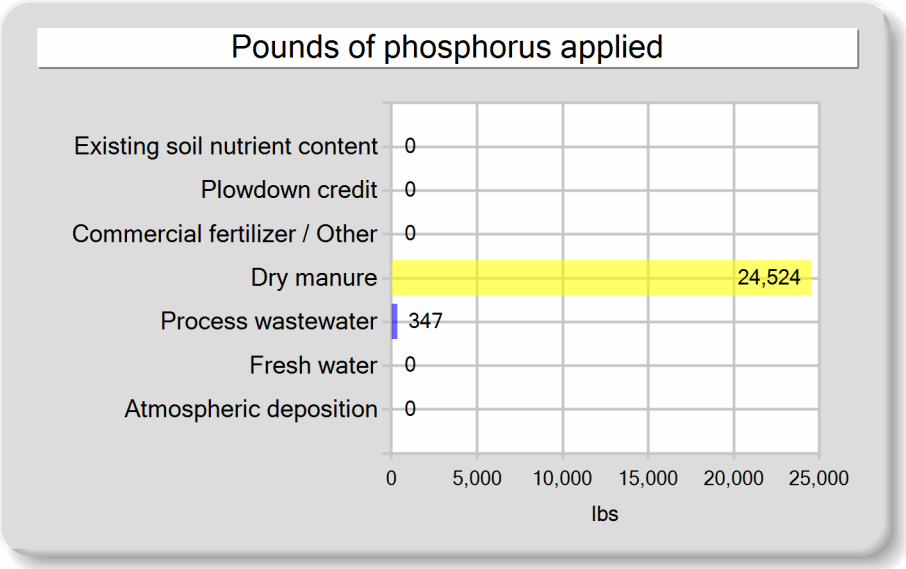
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	63,861.20	24,523.71	88,222.24	0.00
Process wastewater	1,095.87	346.71	4,578.10	17,120.49
Fresh water	0.00	0.00	0.00	54,812.36
Atmospheric deposition	1,820.00	0.00	0.00	0.00
Total nutrients applied	66,777.07	24,870.42	92,800.35	71,932.85
Anticipated crop nutrient removal	52,000.00	9,178.00	43,004.00	0.00
Actual crop nutrient removal	50,975.72	10,181.84	62,992.03	268,245.99
Nutrient balance	15,801.35	14,688.58	29,808.31	-196,313.14
Applied to removed ratio	1.31	2.44	1.47	0.27

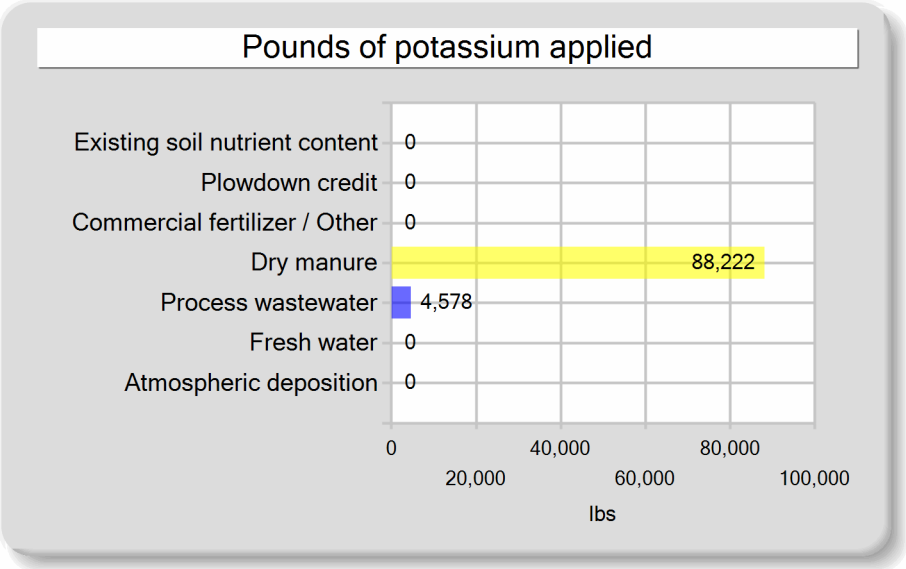
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

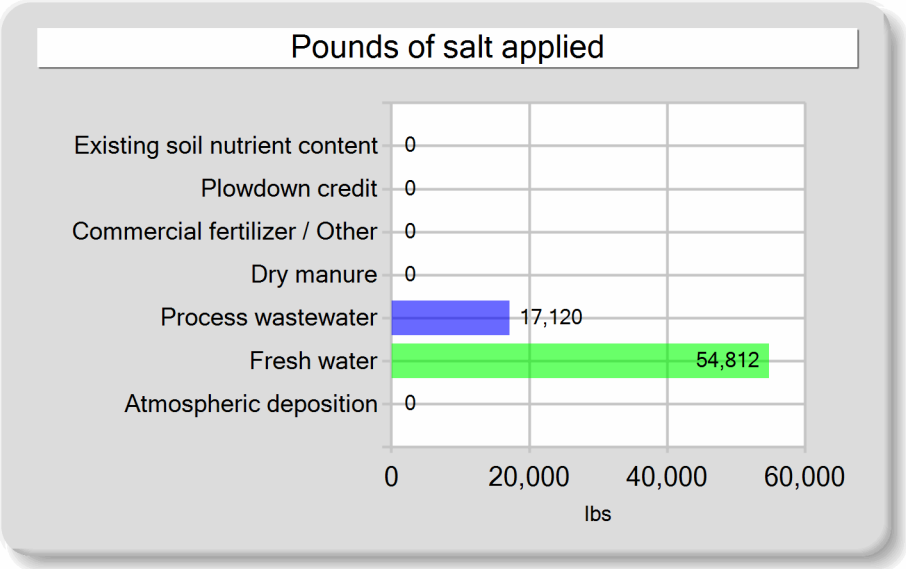


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Dennis Nelson

SIGNATURE OF OWNER OF FACILITY

SIGNATURE OF OPERATOR OF FACILITY

Dennis Nelson

PRINT OR TYPE NAME

SAME AS OWNER

PRINT OR TYPE NAME

6-25-24

DATE

DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Nelson Dairy

Received: 08/03/2023 15:09
Reported: 08/09/2023 09:02

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0525-01	Canal	Ag Water	T. Madruga		08/02/2023 16:00
23H0525-02	ID#987	Ag Water	T. Madruga		08/02/2023 16:00

Default Cooler Temperature on Receipt °C: -1.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Nelson Dairy

Received: 08/03/2023 15:09
Reported: 08/09/2023 09:02

Sample Results

Sample: Canal
23H0525-01 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.03**	mmhos/cm	0.01	1		08/04/23 13:31	SM 2510 B		BEH0245
Electrical Conductivity umhos	**31.1**	umhos/cm	10.0	1		08/04/23 13:31	SM 2510 B		BEH0245
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/03/23 19:42	EPA 300.0		BEH0185
pH	**7.6**	units	1.0	1		08/04/23 13:31	SM 4500-H+	H	BEH0245
Total Filterable Solids (TDS)	**37.0**	mg/L	10.0	1		08/07/23 17:45	SM 2540 C		BEH0226
Temperature	**25.0**	°C	0.0	1		08/04/23 13:31	SM 2510 B		BEH0245
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/08/23 10:03	SM 4500-NH3 C		BEH0286
Total Nitrogen	ND	mg/L	1.00	1		08/08/23 10:03	SM 4500-NH3 C		BEH0286

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Ranch: Nelson Dairy

Received: 08/03/2023 15:09
Reported: 08/09/2023 09:02

Sample Results
(Continued)

Sample: ID#987
23H0525-02 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.46**	mmhos/cm	0.01	1		08/04/23 13:32	SM 2510 B		BEH0245
Electrical Conductivity umhos	**461**	umhos/cm	10.0	1		08/04/23 13:32	SM 2510 B		BEH0245
Nitrate Nitrogen as NO3N	**12.1**	mg/L	0.1	1	10	08/03/23 20:02	EPA 300.0		BEH0185
pH	**6.9**	units	1.0	1		08/04/23 13:32	SM 4500-H+	H	BEH0245
Total Filterable Solids (TDS)	**332**	mg/L	10.0	1		08/07/23 17:45	SM 2540 C		BEH0226
Temperature	**25.0**	°C	0.0	1		08/04/23 13:32	SM 2510 B		BEH0245
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/08/23 10:05	SM 4500-NH3 C		BEH0286
Total Nitrogen	**12.1**	mg/L	1.00	1		08/08/23 10:05	SM 4500-NH3 C		BEH0286

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Ranch: Nelson Dairy

Received: 08/03/2023 15:09
Reported: 08/09/2023 09:02

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0185										
Blank (BEH0185-BLK1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0185-BLK2)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0185-BLK3)					Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0185-BS1)					Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		94.2	90-110		
LCS (BEH0185-BS2)					Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.3	90-110		
Duplicate (BEH0185-DUP1)		**Source: 23H0525-02**			Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	12.2		0.1	mg/L		12.1			0.602	10
Duplicate (BEH0185-DUP2)		**Source: 23H0530-01**			Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	0.07		0.1	mg/L		0.08			12.4	10
Matrix Spike (BEH0185-MS1)		**Source: 23H0525-02**			Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	17.3		0.1	mg/L	5.000	12.1	105	90-110		
Matrix Spike (BEH0185-MS2)		**Source: 23H0530-01**			Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.08	91.7	90-110		
Reference (BEH0185-SRM1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.6	90-110		
Reference (BEH0185-SRM2)					Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.6	90-110		
Reference (BEH0185-SRM3)					Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		98.2	90-110		

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R & J Avila Inc
8119 Ellington Ave
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Ranch: Nelson Dairy

Received: 08/03/2023 15:09
Reported: 08/09/2023 09:02

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0226										
Blank (BEH0226-BLK1)					Prepared: 8/4/2023	Analyzed: 8/7/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEH0226-BS1)					Prepared: 8/4/2023	Analyzed: 8/7/2023				
Total Filterable Solids (TDS)	40.0		10.0	mg/L	2000		2.00	0-200		
Duplicate (BEH0226-DUP1)			**Source: 23H0545-02**		Prepared: 8/4/2023	Analyzed: 8/7/2023				
Total Filterable Solids (TDS)	880		10.0	mg/L		840			4.65	10
Duplicate (BEH0226-DUP2)			**Source: 23H0562-01**		Prepared: 8/4/2023	Analyzed: 8/7/2023				
Total Filterable Solids (TDS)	4020		10.0	mg/L		3820			5.11	10
Reference (BEH0226-SRM1)					Prepared: 8/4/2023	Analyzed: 8/7/2023				
Total Filterable Solids (TDS)	350			mg/L	325.0		108	90-110		
Reference (BEH0226-SRM2)					Prepared: 8/4/2023	Analyzed: 8/7/2023				
Total Filterable Solids (TDS)	500			mg/L	495.0		101	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0245										
Blank (BEH0245-BLK1)					Prepared & Analyzed: 8/4/2023					
pH	5.5		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEH0245-BLK2)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.9		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0245-BLK3)					Prepared & Analyzed: 8/4/2023					
pH	7.2		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH0245-DUP1)		**Source: 23H0525-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.03		0.01	mmhos/cm		0.03			2.28	10
pH	7.1		1.0	units		7.6			5.98	10
Electrical Conductivity umhos	30.4		10.0	umhos/cm		31.1			2.28	10
Duplicate (BEH0245-DUP2)		**Source: 23H0534-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	0.03		0.01	mmhos/cm		0.03			0.587	10
pH	7.6		1.0	units		7.2			5.12	10
Electrical Conductivity umhos	34.2		10.0	umhos/cm		34.0			0.587	10
Reference (BEH0245-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	539			umhos/cm	538.0		100	90-110		
Reference (BEH0245-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.8			units	7.790		100	.7163-101.28		
Reference (BEH0245-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1000			umhos/cm	1000		100	90-110		
Electrical Conductivity umhos	1000			umhos/cm	1000		100	90-110		
Reference (BEH0245-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEH0245-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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Submitted By: Rui & John Avila
Ranch: Nelson Dairy

Received: 08/03/2023 15:09
Reported: 08/09/2023 09:02

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0245 (Continued)										
Reference (BEH0245-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0245-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0245-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0245-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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Ranch: Nelson Dairy

Received: 08/03/2023 15:09
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0286										
Blank (BEH0286-BLK1)					Prepared: 8/7/2023	Analyzed: 8/8/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH0286-BLK2)					Prepared: 8/7/2023	Analyzed: 8/8/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH0286-BS1)					Prepared: 8/7/2023	Analyzed: 8/8/2023				
Kjeldahl Nitrogen (TKN), Total	5.50		1.00	mg/L	5.709		96.3	90-110		
LCS (BEH0286-BS2)					Prepared: 8/7/2023	Analyzed: 8/8/2023				
Kjeldahl Nitrogen (TKN), Total	5.25		1.00	mg/L	5.709		91.9	90-110		
Duplicate (BEH0286-DUP1)		**Source: 23H0322-02**			Prepared: 8/7/2023	Analyzed: 8/8/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BEH0286-DUP2)		**Source: 23H0537-01**			Prepared: 8/7/2023	Analyzed: 8/8/2023				
Kjeldahl Nitrogen (TKN), Total	3.09		1.40	mg/L		2.82			8.99	10
Matrix Spike (BEH0286-MS1)		**Source: 23H0322-02**			Prepared: 8/7/2023	Analyzed: 8/8/2023				
Kjeldahl Nitrogen (TKN), Total	6.68		1.40	mg/L	7.992	ND	83.6	90-110		
Matrix Spike (BEH0286-MS2)		**Source: 23H0537-01**			Prepared: 8/7/2023	Analyzed: 8/8/2023				
Kjeldahl Nitrogen (TKN), Total	7.84		1.40	mg/L	7.992	2.82	62.9	90-110		
Reference (BEH0286-SRM1)					Prepared: 8/7/2023	Analyzed: 8/8/2023				
Kjeldahl Nitrogen (TKN), Total	23.7			mg/L	23.80		99.4	90-110		

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JG

08/03/23 15:09 23H0525

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 2 No. of Bottles: 4

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Nelson Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

( ) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(X) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Canal	8/2/2023	1600	-1.4	
2	ID#987	8/2/2023	1600	-1.2	
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/2/2023 @ 1600	8/3/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	8/3/2023 @ 900	8-3-23 2:30
Third				
Fourth	[illegible]	DLI	8/3/23 15:09	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs _____ $ _____ In
Miles _____ $ _____ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/03/23 15:09 23H0525

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)	1									
250 mL H_2SO_4 (Yellow)	1	1								
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

JG 8/3/23

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Nelson Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:06

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0896-01	DN #3	Ag Water	T. Madruga		11/15/2023 15:00
23K0896-02	ND #1	Ag Water	T. Madruga		11/15/2023 15:30
23K0896-03	ND #2	Ag Water	T. Madruga		11/15/2023 16:00

Default Cooler Temperature on Receipt °C: 2.3
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: Nelson Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:06

Sample Results

Sample: DN #3
23K0896-01 (Water)

Sampled: 11/15/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.51**	mmhos/cm	0.01	1		11/20/23 13:40	SM 2510 B		BEK0616
Electrical Conductivity umhos	**509**	umhos/cm	10.0	1		11/20/23 13:40	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 15:00	Field		BEK0556
Nitrate Nitrogen as NO3N	**15.9**	mg/L	0.1	1	10	11/16/23 20:31	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/20/23 13:40	SM 4500-H+	H	BEK0616
pH	**7.9**	units	1.0	1		11/20/23 13:40	SM 4500-H+	H	BEK0616

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LABORATORY INC

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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Nelson Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:06

Sample Results
(Continued)

Sample: ND #1
23K0896-02 (Water)

Sampled: 11/15/2023 15:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.47**	mmhos/cm	0.01	1		11/20/23 13:42	SM 2510 B		BEK0616
Electrical Conductivity umhos	**473**	umhos/cm	10.0	1		11/20/23 13:42	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 15:30	Field		BEK0556
Nitrate Nitrogen as NO3N	**5.8**	mg/L	0.1	1	10	11/16/23 20:52	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/20/23 13:42	SM 4500-H+	H	BEK0616
pH	**7.9**	units	1.0	1		11/20/23 13:42	SM 4500-H+	H	BEK0616

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Nelson Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:06

Sample Results
(Continued)

Sample: ND #2
23K0896-03 (Water)

Sampled: 11/15/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.47**	mmhos/cm	0.01	1		11/20/23 13:43	SM 2510 B		BEK0616
Electrical Conductivity umhos	**474**	umhos/cm	10.0	1		11/20/23 13:43	SM 2510 B		BEK0616
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 16:00	Field		BEK0556
Nitrate Nitrogen as NO3N	**5.9**	mg/L	0.1	1	10	11/16/23 21:12	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/20/23 13:43	SM 4500-H+	H	BEK0616
pH	**7.8**	units	1.0	1		11/20/23 13:43	SM 4500-H+	H	BEK0616

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DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Nelson Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:06

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0573										
Blank (BEK0573-BLK1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK2)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK4)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0573-BS1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEK0573-BS2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
LCS (BEK0573-BS3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEK0573-DUP1)		**Source: 23K0896-02**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.8		0.1	mg/L		5.8			0.173	10
Duplicate (BEK0573-DUP2)		**Source: 23K0901-01**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	0.3		0.1	mg/L		0.3			0.629	10
Duplicate (BEK0573-DUP3)		**Source: 23K0935-02**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	1.5		0.1	mg/L		1.5			0.195	10
Matrix Spike (BEK0573-MS1)		**Source: 23K0896-02**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	11.0		0.1	mg/L	5.000	5.8	104	90-110		
Matrix Spike (BEK0573-MS2)		**Source: 23K0901-01**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.3	97.1	90-110		
Matrix Spike (BEK0573-MS3)		**Source: 23K0935-02**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.5	106	90-110		
Reference (BEK0573-SRM1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEK0573-SRM2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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LABORATORY INC

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Ranch: Nelson Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:06

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0573 (Continued)										
Reference (BEK0573-SRM3)					Prepared: 11/16/2023	Analyzed: 11/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0573-SRM4)					Prepared: 11/16/2023	Analyzed: 11/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Ranch: Nelson Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:06

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0616										
Blank (BEK0616-BLK1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Blank (BEK0616-BLK2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.2		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0616-BLK3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	6.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEK0616-DUP1)		**Source: 23K0896-03**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	0.47		0.01	mmhos/cm		0.47			0.423	10
Electrical Conductivity umhos	472		10.0	umhos/cm		474			0.423	10
pH	7.8		1.0	units		7.8			0.255	10
Duplicate (BEK0616-DUP2)		**Source: 23K0899-01**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1.65		0.01	mmhos/cm		1.64			0.298	10
Electrical Conductivity umhos	1650		10.0	umhos/cm		1640			0.298	10
pH	7.0		1.0	units		7.0			0.00	10
Reference (BEK0616-SRM1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	561			umhos/cm	538.0		104	90-110		
Reference (BEK0616-SRM2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	7.6			units	7.520		101	67021-101.32		
Reference (BEK0616-SRM3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0616-SRM4)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEK0616-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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Received: 11/16/2023 15:35
Reported: 11/22/2023 09:06

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0616 (Continued)										
Reference (BEK0616-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEK0616-SRM6)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0616-SRM7)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0616-SRM8)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/16/23 15:35 23K0896

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 3 No. of Bottles: 3

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Nelson Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DN #3	11/15/2023	1500	2.3	ND
2	ND #1	11/15/2023	1530	2.2	ND
3	ND #2	11/15/2023	1600	2.2	ND
4					
5					
6					
7	IR Thermometer SN: 221314357				
8	Correction Factor: 0°C				
9	Calibration Due: 12/22/2023				
10	Location: Laboratory				

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/15/2023 @ 1600	11/16/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/16/2023 @9am	11-16-23 3:25
Third		DLI	11-16-23 15:35	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:35 23K0896

Shipping In ... Fed Ex □ UPS □ GSO □ Picked-Up ☑ Walk In □ DLI Sampler □
Ontrac □ PMS □ USPS □ Other □ _____

Shipping Container: Ice Chest ☑ None □ Box □

Refrigerant: Wet Ice: ☑ Blue Ice: □ None: □

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										